Department of Computer Science and Engineering

UIT,RGPV ,Bhopal

List of Assignment - 2

Web Engineering (Sec A and Sec B)

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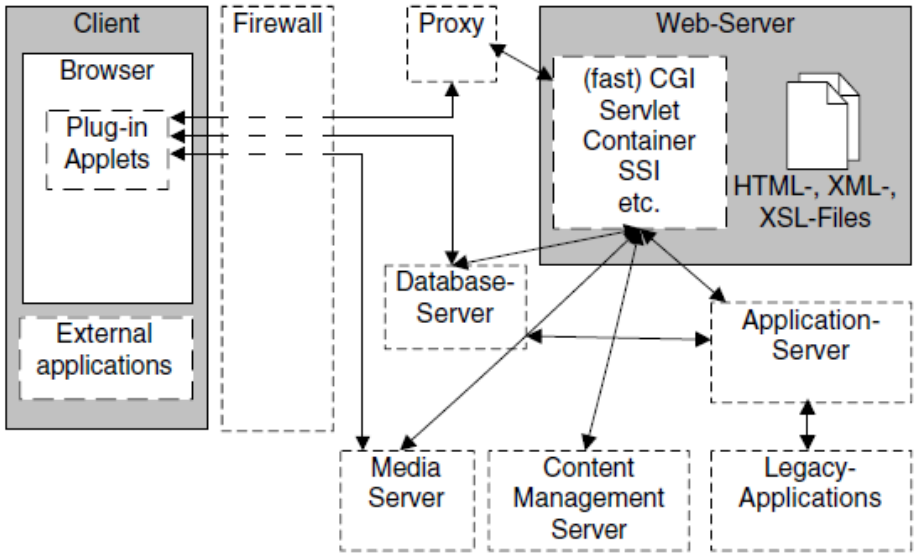


Figure.1 Basic components of Web application

Evaluate above fig.1 carefully and describe the role of Content management server and Legacy application

2. Discuss the problems and restrictions in Integrated Web Design ;Also Discuss the Search fundamentals and their strategies.

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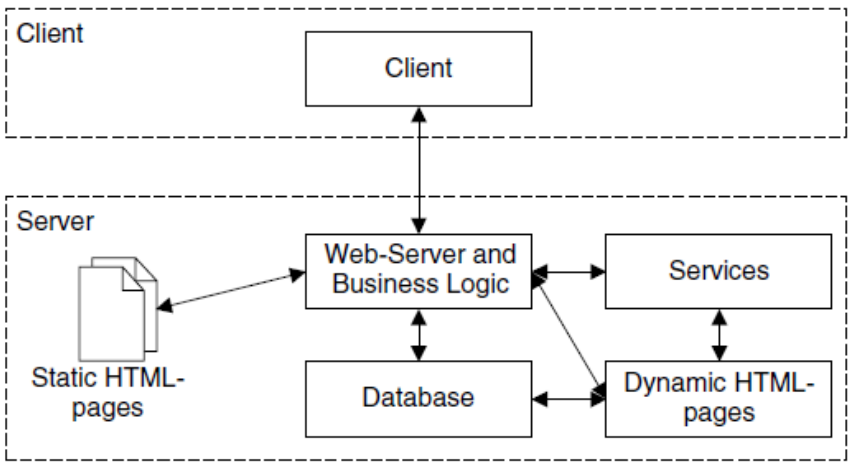


Figure 2. Architecture of Web applications

Interpret and explain each component working as listed above fig.2

4

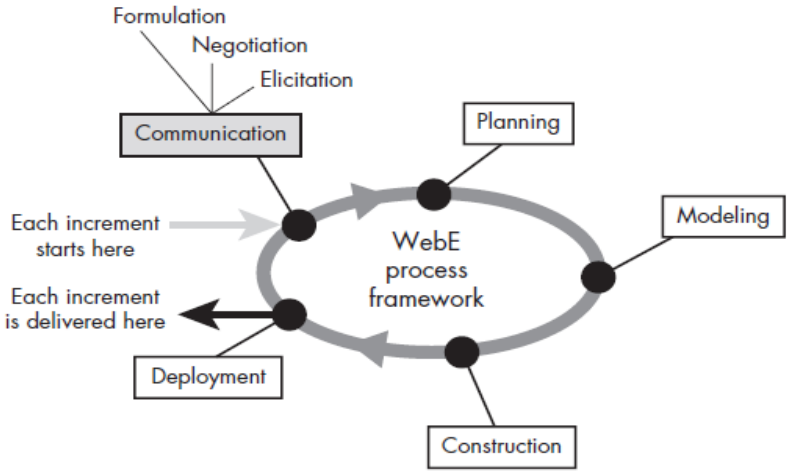


Figure. 3 The WebE process Framework activity

What Techniques Can You Use for Communication and Why? Justify your answer with referred above fig. 3?

5. What is Email (WebApp or Service)? Justify your answer.

6.

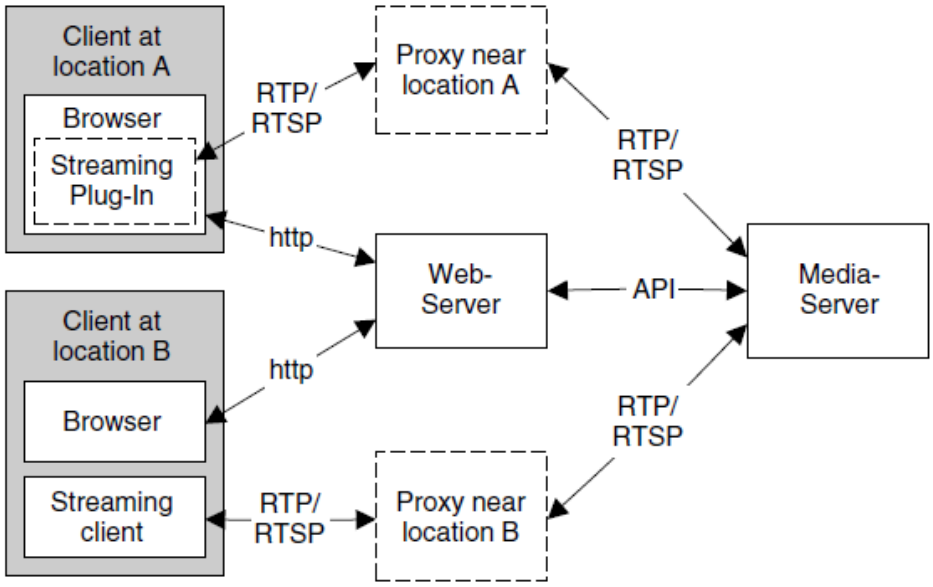


Figure. 4 Streaming media architecture using point-to-point connections

You can identify two distinct client locations for multimedia data streaming applications. First, identify existing content available on-demand, e.g., video-on-demand, and compare this model with broadcasting live content to many users ; Compare Webcasting different requirements based on architectures fig. 4 and Identify the significant causes for bandwidth and server load problems.

7. How does RE for Web engineering differ from RE for conventional software

systems? and what do you mean by Twin Peaks model? Explain in brief.

8. Interactive Web Applications must be able to distinguish requests by multiple simultaneous users and identify related requests coming from the same user. Which term is used to define such a sequence of related HTTP requests between a specific user and a server within a specific time window? Discuss with examples.

9.

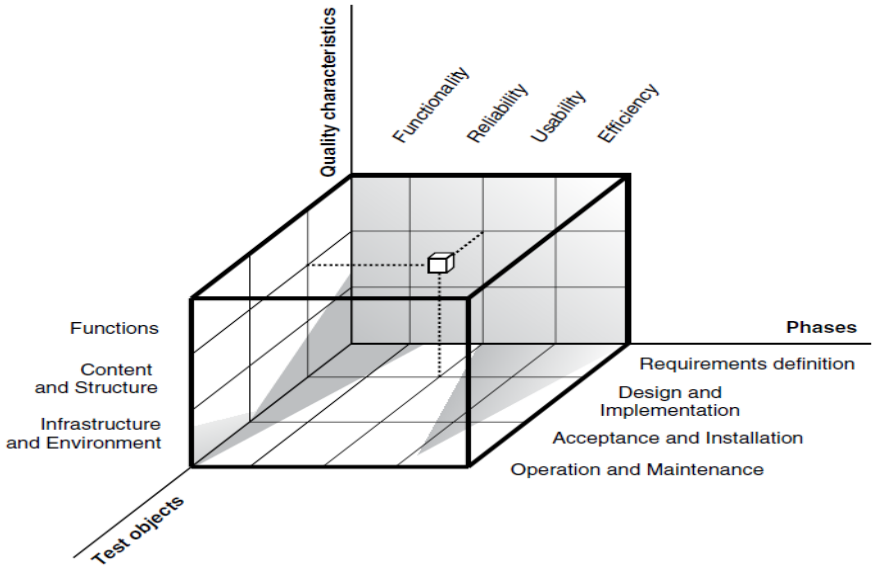


Figure. 5 Test scheme for Web applications

If you join these three dimensions – quality characteristics, test objects, and phases – the result can be visualized as a three-dimensional cube as shown in fig. 5; Explain each tests outlines of cube contains. Also discuss What Types of Errors Occur Within a WebApp Environment?

10. Explain why large WebApps incorporate searching mechanisms? Discuss the Steps of Searching Mechanisms Lead to Problems? Also Explain How Is the Relationship Management Methodology (RMM) Model Used for Navigation Design?

11. Explain the necessity of Interface design for WebApps. How Are Interface Objects and Actions Translated into a Layout?

12. Why Is WebML Appropriate for Workflow Modeling? Justify your answers. Also explain how is design focus used for identify patterns?

13. List the merits and demerits of Link popularity and PageRank

14. What is Usability of Web applications? Explain in brief ; Draw and discuss Layered usability model.

15. What do you mean by Web Caching and Content Pre-fetching also discuss the

fundamentals of the Semantic Web

16. XML (eXtensible Markup Language)document is well-formed, there are general rules for the syntax of XML

Correct the following XML well-formedness rules

(i) <A><B></A></B> [All tags have to appear in pairs on the same nesting depth. In addition, there are empty tags (<B/>)].
(ii) <A> [Tags are uppercase- and lowercase-sensitive].
(iii) <A elem = 5/> [All attributes must be enclosed in quotation marks].
(iv) <A B></A B> [Tag names have to comply with the rules for element names. For example, blanks and < or > are invalid].

17.

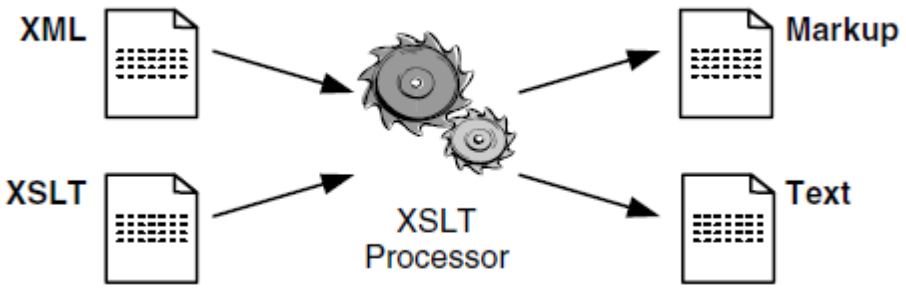


Figure. 6 Schematic view of an XSLT processor works

The XSLT language elements can be grouped in four distinct categories: classify them and also explain XSLT processor working as above shown in fig.6.

18. A Server-Side Include (SSI) represents a simple mechanism allowing the creation of HTML pages, composed of several text fragments, on the fly. This mechanism is implemented by a pre-processor for HTML pages, which is normally integrated in modern Web servers. An HTML page that uses SSI is forwarded to the pre-processor when a Web browser requests it. The pre-processor's output is then delivered by the server. The server generally identifies such pages by a special file extension (normally shtml). The commands used to control the pre-processor are embedded in SGML comments and have the following form:

<!--#command attribute=value ... attribute=value -->

The value of command specifies the command type, which is parameterized by an attribute-value list. Explain what types of commands are supported by SSI; Justify your answer.

19.

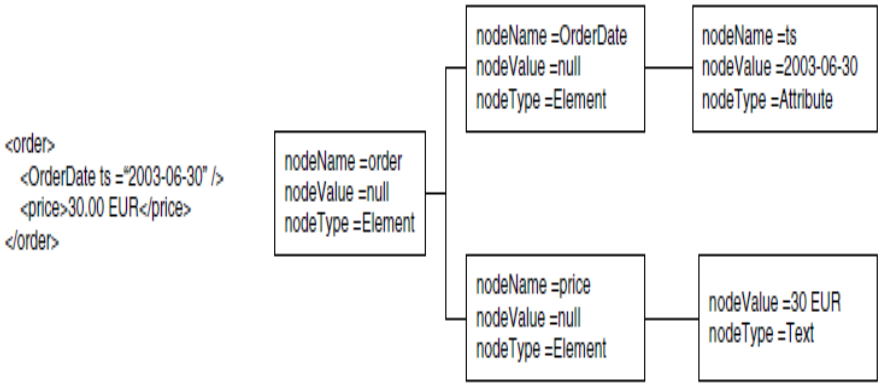


Figure.7 XML example of the DOM structure for a fragment of the "purchase order"

The above Fig.7 shows a Document Object Model (DOM) of an object-oriented view on XML documents; illustrate the following:
(i) Which parses and structure of an XML document are instantiates an object tree.
(ii) Write the benefit of this method is to access the nodes in an object-oriented ways.
(iii) The drawback of this approach.

20.

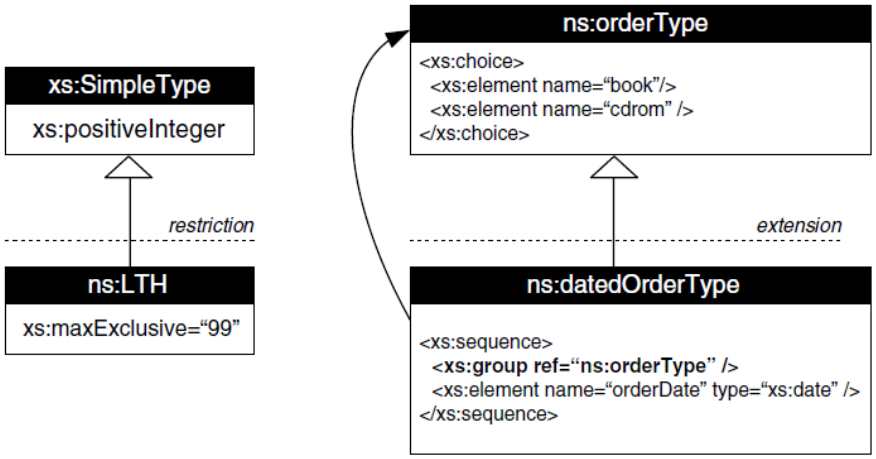


Figure.8 User-defined types

An XML schema can be used to describe various pre-defined data types, such as string, byte, decimal, or date. In addition, they let one define facets which support user-defined data types similar to templates. Let's assume, for example, that all valid ISBN attributes of the XML element book in the "purchase order" example must follow the notation for ISBNs. illustrate the above fig.8 explain uses of a facet, and describe the combined numbers and dashes with the pattern N\-NNNN\-NNNN\-N.

21.

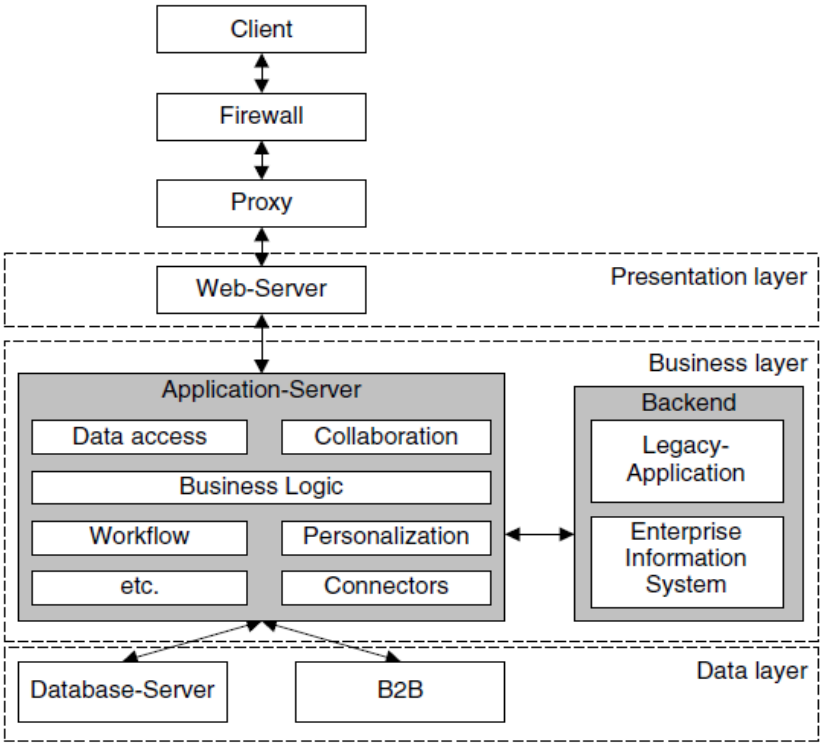


Figure.9 An n-layer architecture for Web applications

Observed fig.9 carefully and identify which one is used to integrate external systems, e.g., business partner systems, or to integrate legacy applications and enterprise information systems. Explain in brief .

22. What do you mean by the SSL Handshake Protocol and Digital Signature; Explain in brief.

23. Design the navigation scheme : Short answer the following
(i)What navigation links and nodes are required?
(ii)What navigation conventions and aids are to be used?
(iii) Do navigation mechanisms correspond to the interface requirements and design?
(iv)Has navigation been optimized for different user categories?
(v) Do navigation semantics agree with each usage scenario?

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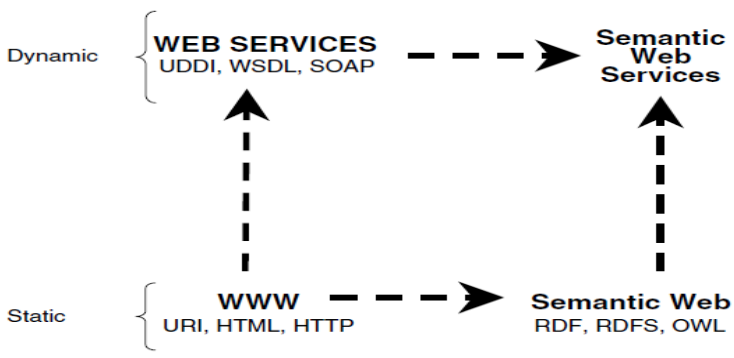


Figure .10 Technical details on Web Services

For example, "A service that sells airline tickets between Vienna and Amsterdam and that accepts payment by Master Card." Currently, the user must perform this task, first using a search engine to find a service and then either reading the Web site associated with that service or executing the service to see and check whether it adheres to the specified task true or not before doing it. Above fig.10 identifies the different E-commerce services and who can specify the information necessary for web service discovery as computer-interpretable semantic markup at the service websites, a service registry or an (ontology-enhanced) search engine. How it can automatically locate appropriate services to do it, described them.

25. E-commerce, cashless payment systems ; At present, credit cards are a very widespread payment system used for Internet transactions. Customers transmit their name, credit card number, and expiration date to the vendor or a trusted intermediary payment service. In order to this write the details working procedures of Electronic Payment Systems and Credit Card Security.

26. Define Hyper Text Markup Language , why this is the most popular standard markup language for documents designed to be displayed in a web browser. Also explain their technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript.

27. Element example

<**head**>
<**title**>The Title</**title**>
<**link** rel="stylesheet" href="stylebyjimbowales.css" /> *<!-- Imports Stylesheets -->*
</**head**>

HTML documents imply a structure of nested HTML elements. These are indicated in the document by HTML tags, enclosed in angle brackets thus define given example in brief.

28. HTTP cookies are known as web cookies, Internet cookies, browser cookies, or simply cookies; It is a small blocks of data created by a web server while a user is browsing a website and placed on the user's computer or other device by the user's web browser.

State your answer

(i) Is Cookies are placed on the device used to access a website, and more than one cookie may be placed on a user's device during a session?

(ii) Is it can serve useful and essential functions on the web?

(iii) Is Authentication cookies commonly used by web servers to authenticate?

(iv) Is Tracking cookies, and especially third-party tracking cookies, commonly used as ways to compile long-term records of individuals?

(v) Is a cookie wall pop up on a website informing the user of the website's cookie usage?